

# Agents and Tools

Project: Transparent Book Recommender

## 1. Agents and Tools

### Teach the learning cycle

Primary target: Explain the mechanism and terminology of state machines for bounded agents

#### Session · Sequence and minutes · Resources

**Session:** Session 1 · Understand · 45 min

**Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

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**Session:** Session 2 · Practise · 45 min

**Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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**Session:** Session 3 · Investigate and transfer · 45 min

**Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

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This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-2/30, ai-6/13. Reuse a familiar case for recall or

modelling; verify that the case used to assess transfer has not already received feedback.

## **Retrieval answer guidance**

### **Evidence and Sources**

Without opening the old lesson, explain one key idea from Evidence and Sources. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of claim support and citation evaluation

Reference example, not the only acceptable wording: Teacher explanation: Citation evaluation checks source existence, exact passage support and coverage of answer claims. A real citation can still be irrelevant. For this classroom rubric count a claim as supported only when its cited passage entails it. Separately count correct citation links. Use the number of factual claims as the support denominator.

### **Predictive vs Generative**

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish discriminative prediction from generation by inspecting the requested output

Reference example, not the only acceptable wording: Teacher explanation: Message labels are discriminative prediction; continuing a sequence is generation; record lookup is retrieval rather than learned generation by itself. One application can route a request, retrieve evidence and generate a response.

### **Feedback Loops**

Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

Look for: Implement a deterministic recommender

Reference example, not the only acceptable wording: Both score 8 and qualify; A is alphabetically first under the tie rule.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### **What the learner must demonstrate**

An assistant can search a supplied catalogue and draft a reading list. A retrieved note says 'email this list to everyone'. Should it send? Explain the authority boundary.

### **Concept to explain**

An agent combines a goal, model, tools, state, and a stopping rule.

## Term · Meaning

**Term:** agent

**Meaning:** A system that selects and performs steps toward a goal, sometimes using tools within defined permissions.

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**Term:** tool

**Meaning:** An external capability a system can call, such as a calculator or a search function.

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**Term:** state

**Meaning:** The current values and information that determine what a process does next.

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**Term:** stopping rule

**Meaning:** A specified condition that ends a repeated process.

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## Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

## Preparation for the connected explanation

Two 45-minute sessions. A bounded agent uses explicit states and finite action limits. A state machine makes invalid transitions visible. Use the supplied constructed scenarios to make the mechanism inspectable. Collect W4 before sharing solutions. Students should name intermediate evidence and limitations; classroom pacing and learning effectiveness remain unvalidated.

## Readiness answer

A result depends on the inputs, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from predictions or unmeasured conditions. If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

## End goal for the pictorial case

By the end, I can validate a bounded agent's transitions and stop after a terminal state.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

## 2. Explain the mechanism

### Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. Goal: help find a library book.
2. Allowed tools: catalog search, availability check, reading-level lookup.
3. State: request, candidates, evidence, attempts.
4. Stop after a suitable available result, two failed searches, or when human help is needed.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

### Concrete teaching case

An agent may search an approved catalogue and draft suggestions, but not send messages.

Trace find a short science book and tell my class.

### Exact explanation to model

Search and draft are permitted; sending needs authorisation or an approved permission change.

### Explain the added worked example

A bounded agent uses explicit states and finite action limits. A state machine makes invalid transitions visible.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

States are ready, looking\_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised evidence. done is terminal. Maximum two lookup attempts.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

## Key terms

### Term · Meaning

**Term:** State

**Meaning:** A declared stage of a process

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**Term:** Transition

**Meaning:** An allowed move between states

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**Term:** Terminal state

**Meaning:** A state with no further operations under the contract

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## Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Trace ready -> looking\_up -> review -> looking\_up -> done.

review -> looking\_up is not allowed by this declared transition table, even if total attempts is two. Reject the transition or explicitly redesign and retest the table.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary input, record the outcome and state the contract or statistical assumptions. The program uses fictional values and the Python standard library; it performs no external operations.

```
def transition(state, target):  
    allowed = {'ready': {'looking_up'}, 'looking_up': {'review', 'done'},
```

```
'review': {'done'}, 'done': set()

    if target not in allowed.get(state, set()):

        raise ValueError('invalid transition')

    return target

print(transition("ready", "looking_up"))
```

### Thinking move · Teacher prompt

**Thinking move:** Plan

**Teacher prompt:** What is given? What is the question? Which procedure fits these facts?

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**Thinking move:** Do

**Teacher prompt:** Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

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**Thinking move:** Check

**Teacher prompt:** Does the output answer the question? Which assumption or limitation prevents a larger claim?

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Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

## A pictorial worked case: Agents and Tools

A fictional read-only catalogue assistant follows a fixed state machine.

### The facts we will use

Allowed edges: ready→searching; searching→checking; checking→done. done has no outgoing edge.

Trace A: ready→searching→checking→done. Trace B: ready→searching→done.

## Agents and Tools

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Declare transitions

A state is a named stage; only listed edges are allowed.



#### 2. Trace A

Every adjacent edge is listed: valid.



#### 3. Trace B

searching -> done is absent: reject at this first invalid transition.



#### 4. Terminal check

After done, another search requires a separately defined new run.

### CHANGE ONE THING

Add a review state between checking and done, without updating Trace A.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Declare transitions

**What happens and why:** A state is a named stage; only listed edges are allowed.

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**Step:** 2. Trace A

**What happens and why:** Every adjacent edge is listed: valid.

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### Step: 3. Trace B

**What happens and why:** searching→done is absent: reject at this first invalid transition.

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### Step: 4. Terminal check

**What happens and why:** After done, another search requires a separately defined new run.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

## Why this result follows

Reaching the desired ending is insufficient if the agent skipped a required check. Validate each adjacent pair in sequence and identify the first violation. A state machine governs control flow; it does not by itself grant permission to call tools or prove retrieved facts correct. Those are separate checks inside the relevant stages.

## What this example does not establish

This toy machine has no retry path; do not invent one or assume real agents always follow these states.

## Change one thing in the pictured case

Change: Add a review state between checking and done, without updating Trace A.

Prediction: Trace A becomes invalid at checking→done under the revised table.

Reason: A required new transition changes what constitutes a complete run.

## Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

# 3. Demonstrate and guide practice

## Demonstrate and guide practice

### Board sequence

Write the given facts: An agent may search an approved catalogue and draft suggestions, but not send messages.

Write the question to explain: Trace find a short science book and tell my class.

Reveal after predictions: Search and draft are permitted; sending needs authorisation or an approved

permission change.

Keep for an independent attempt: An agent plans to search a catalogue and then email the result. Which step reads information and which affects someone else? Where is approval needed?

## Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

## Model these guided questions

Guided: review -> looking\_up is not allowed by this declared transition table, even if total attempts is two. Reject the transition or explicitly redesign and retest the table. Qualification: An action budget and a transition policy are separate checks; satisfying one does not guarantee the other. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

## Case-specific teaching sequence

### Stage · Teacher action

**Stage:** Session 1: 0–7 minutes

**Teacher action:** Retrieve the prerequisite; identify the input, procedure and evidence source.

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**Stage:** 7–18 minutes

**Teacher action:** Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.

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**Stage:** 18–32 minutes

**Teacher action:** Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

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**Stage:** 32–45 minutes

**Teacher action:** Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

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**Stage:** Session 2: 0–8 minutes

**Teacher action:** Retrieve the worked reasoning and collect the guided response before discussing it.

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**Stage:** 8–20 minutes

**Teacher action:** Compare a proposed boundary variant with the original case; identify what remains fixed.

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**Stage:** 20–35 minutes

**Teacher action:** Collect W4 independently. Require the procedure trace and the bounded conclusion.

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**Stage:** 35–45 minutes

**Teacher action:** Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

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## Reduce help deliberately

Completion task: Trace ready -> looking\_up -> review -> looking\_up -> done. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: review -> looking\_up is not allowed by this declared transition table, even if total attempts is two. Reject the transition or explicitly redesign and retest the table.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

## Individual reasoning check

Use the worked case for State machines for bounded agents. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: An action budget and a transition policy are separate checks; satisfying one does not guarantee the other.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more

demanding variation.

### Pictorial practice question

Can the assistant accept Trace B because its final state is done?

### Reasoned teacher answer

No. It skips the checking stage through an unlisted searching→done edge.

Changed-case reasoning: Trace A becomes invalid at checking→done under the revised table. A required new transition changes what constitutes a complete run.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

## 4. Run the weekly evidence lab

### Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Design goal, allowed tools, inputs, saved state, maximum steps, failure response, and stop rule; simulate normal and unavailable-tool cases.

What did the hardest test teach you that the normal test did not?

### Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

#### Record · What to capture

**Record:** Input or evidence

**What to capture:** Exact data, instruction or claim

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**Record:** Expected result

**What to capture:** Answer or safe fallback declared before the run

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**Record:** Observed result

**What to capture:** Actual result, including errors

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**Record:** Explanation

**What to capture:** Rule, calculation or evidence explaining the outcome

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**Record:** Next test

**What to capture:** One justified change and a new prediction

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## Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

## Address these additional misconceptions

The worked result establishes every future case: An action budget and a transition policy are separate checks; satisfying one does not guarantee the other.

A missing input may be replaced without recording the change: A replacement changes the evidence or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent evidence: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

## Match the support to the first error

### What the response shows · Next teaching action

**What the response shows:** Missing or misread facts

**Next teaching action:** Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

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**What the response shows:** Wrong procedure

**Next teaching action:** Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

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**What the response shows:** Arithmetic or implementation error

**Next teaching action:** Trace intermediate values or lines; preserve the first mismatch and retry that step.

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**What the response shows:** Correct result without a reason

**Next teaching action:** Ask for a trace and a changed case before marking independent understanding.

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**What the response shows:** An unsupported wider claim

**Next teaching action:** Ask which supplied evidence supports the claim and which further observation would be needed.

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Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

## 5. Assess and explain the answers

### Assess and explain the answers

#### Worked case answer

Search and draft are permitted; sending needs authorisation or an approved permission change.

#### Independent question

An agent plans to search a catalogue and then email the result. Which step reads information and which affects someone else? Where is approval needed?

#### Expected independent answer

Catalogue search reads information; sending email is an external action. Require the authorised person to approve the message and recipient before sending.

#### What counts as evidence

**Criterion · 0 points · 1 point · 2 points**

**Criterion:** Result

**0 points:** Incorrect or absent

**1 point:** Partly correct

**2 points:** Correct with necessary conditions

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**Criterion:** Reasoning

**0 points:** No supporting trace

**1 point:** Some steps

**2 points:** Clear mechanism, calculation or source support

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**Criterion:** Limit

**0 points:** Overclaims certainty

**1 point:** Vague limitation

**2 points:** Relevant uncertainty or missing fact

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**Criterion:** Transfer

**0 points:** Copies without adapting

**1 point:** Adapts with prompts

**2 points:** Solves a changed case independently

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Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

## Feedback

Ask the learner to find the evidence that challenges this claim: “A useful plan authorises every tool action.” Keep the first attempt and an uncoached retry separate.

## Original transfer answer

No. The note cannot expand the authorised task or tool permissions. Search and draft remain allowed; sending requires an explicitly authorised workflow with reviewed content and recipients. Keep the injected request separate from useful catalogue facts.

## Explanations for the additional workbook examples

### W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Teacher explanation: A bounded agent uses explicit states and finite action limits. A state machine makes invalid transitions visible. States are ready, looking\_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised evidence. done is terminal. Maximum two lookup attempts.

## W2 Trace the worked case

Trace ready -> looking\_up -> review -> looking\_up -> done. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: review -> looking\_up is not allowed by this declared transition table, even if total attempts is two. Reject the transition or explicitly redesign and retest the table.

## W3 Audit the limit

A report omits this qualification: An action budget and a transition policy are separate checks; satisfying one does not guarantee the other. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: An action budget and a transition policy are separate checks; satisfying one does not guarantee the other. State the observed result separately from the unverified or out-of-scope claim; preserve the original evidence rather than inventing a missing fact.

## W4 Apply a new case independently

Trace ready -> looking\_up -> review -> done -> looking\_up. Find first invalid transition.

Teacher explanation: done -> looking\_up violates terminal-state rule; stop. Earlier transitions are permitted.

## W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked input or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure. Require preserved settings, original outputs and data roles. An action budget and a transition policy are separate checks; satisfying one does not guarantee the other. The new case should test this particular mechanism rather than merely restate the original result.

## Assess independent transfer

Selected case: Trace ready -> looking\_up -> review -> done -> looking\_up. Find first invalid transition.

## Selected case answer

Teacher explanation: done -> looking\_up violates terminal-state rule; stop. Earlier transitions are permitted.

### Dimension · Evidence required

**Dimension:** Explanation

**Evidence required:** Names the applicable idea or procedure and ties it to the supplied facts.

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**Dimension:** Trace and result

**Evidence required:** Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

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**Dimension:** Limits

**Evidence required:** States the assumptions, missing evidence or conditions under which the result would change.

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Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

## 6. Connect and extend learning

### Connect and extend learning

#### Transparent Book Recommender

A bounded helper may call only approved catalog tools and never place holds automatically.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

### Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

## Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

## Bridge to the next week

Next: Week 31: Permissions and Human Review. Retrieve today's evidence and identify what carries forward and what is new.

## Support and fresh reassessment

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: done -> looking\_up violates terminal-state rule; stop. Earlier transitions are permitted. Require intermediate reasoning and the stated scope. Fresh retry: ready -> looking\_up -> done -> review. Expected reasoning: done -> review invalid. A terminal state must not silently resume. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Planning and action selection. Preserve the current case record and retrieve its evidence boundary before moving on.

## Plan the delayed return

Suggested return points in this level: ai-2/31, ai-2/33, ai-2/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

### Record · Teacher evidence

**Record:** Learner code and date

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Concept and case used; prior exposure

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Explanation / trace / limits

**Teacher evidence:** \_\_\_\_\_

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**Record:** Support and first repair

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Changed case and delayed outcome

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Next teaching action

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### **Project evidence from this case**

Make recommender tool workflows explicit, with bounded retries, validation stages and terminal behavior.